STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 12A

Description

12.Write R program to do the following tests with the sample data and visualize the results graphically.

a) χ2-test

Procedure and Overvation:

chi square test is a statistical method to determine if 2 categorical variables have a significant correlation between them then those variables should be from same population and there should be categorical like yes/no, male/female, Red/Green

Example:

We can build a dataset with observations on people icecream buying pattern and try to correlate the gender of a person with the flavor of the ice cream they prefer if correlation is found we can plan for appropriate stock of flavors by knowing the no. of gender of people visiting.

SYNTAX: chisqr.test(data)

data is the data in the form of a table containing the count value of the variables in the observation.

SOURCE CODE

```
# Loading the Mass library.
library("MASS")
print(str(Cars93))
# Creating a data frame from the main data set.
car_data<- data.frame(Cars93$AirBags, Cars93$Type)
# Creating a table with the needed variables.
car_data = table(Cars93$AirBags, Cars93$Type)
print(car_data)
# Performing the Chi-Square test.
print(chisq.test(car_data))
```

NULL

```
                      Compact Large Midsize Small Sporty Van
  Driver & Passenger        2     4       7     0      3   0
  Driver only               9     7      11     5      8   3
  None                      5     0       4    16      3   6

        Pearson's Chi-squared test

data:  car_data
X-squared = 33.001, df = 10, p-value = 0.0002723
```

VIVA RESPONSES

Q)

Q)

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